

Aritra Mondal

Data Scientist | Machine Learning | Predictive Analytics

Production ML (Amadeus) · Applied MSc Bac+5 (Summa Cum Laude) · Immediate Availability (CDI Cadre)

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Location: Paris / Biarritz, France · Immediate Relocation Ready (Hybrid / Onsite) **Work Rights:** Titre de séjour (Full Legal Right to Work across France & EU) · **Notice:** 0 Days

PROFESSIONAL SUMMARY

Production-minded **Data Scientist** with proven enterprise experience scaling distributed machine learning and data pipelines at **Amadeus**. Specialized in **time-series demand forecasting**, distributed compute (**Spark, Scala, AWS EMR**), and automated **MLOps CI/CD**. Holds an **Applied MSc in Data Science & AI (Summa Cum Laude, Bac+5)** and **BSc (Hons) Computer Security**. Immediately available for CDI roles across France.

WORK EXPERIENCE

Data Scientist (Apprentice) — Amadeus

Aug 2023 – Sep 2024

Sophia Antipolis, France · Hybrid (Airline Revenue Management & Demand Forecasting)

- Engineered and deployed production **machine learning pipelines** for airline demand forecasting on distributed **AWS EMR** clusters and **Amazon S3**.
- Accelerated ETL batch processing performance by **35%** by migrating legacy PySpark workflows to native **Scala**, eliminating JVM-Python IPC overhead.
- Constructed automated **CI/CD pipelines** using **Jenkins** and **Bash**, standardizing reproducible testing artifacts and reducing manual intervention by **50%**.
- Engineered scalable **feature pipelines** over high-volume structured **Parquet** data lakes, feeding high-throughput downstream predictive models.

Android / Software Developer (Intern) — Jungleworks

Nov 2020 – Apr 2021

100% Remote · Mobile Software Architecture & API Integrations

- Developed native Android modules using **Java / Kotlin**, integrating asynchronous **REST APIs** with offline caching and defensive error handling.
- Instituted structured unit testing and debugging workflows that eliminated core crash vectors and elevated app store stability.

EDUCATION & DEGREES

Applied MSc in Data Science & AI (Bac+5 Equivalent)

2022 – 2024

Data ScienceTech Institute (DSTI) · *Sophia Antipolis & Paris, France*

Honors: First Class / *Summa Cum Laude — Félicitations du Jury*. Core: Machine Learning, Deep Learning (PyTorch), Big Data (Spark, Scala), MLOps (Docker, Jenkins), Cloud (AWS, Azure), Neo4j. Completed in alternance at Amadeus.

BSc (Hons) Computer Security (Bac+3 Equivalent)

2017 – 2020

Northumbria University · *Newcastle upon Tyne, United Kingdom*

Core: Network Security (TCP/IP, UDP, ICMP), Linux Internals, Cryptography, Database Systems & SQL, Ethical Hacking. Thesis: SNORT IDS heuristic false-positive reduction.

Advanced Diploma in Information Technology (Bac+2)

2015 – 2017

Management Development Institute of Singapore (MDIS) · *Singapore*

Object-Oriented Programming (Java, C++), Relational Databases & SQL, Distributed Systems.

CORE COMPETENCIES

MACHINE LEARNING & AI

Python · Scikit-learn · XGBoost · LightGBM · Time-Series · PyTorch / LSTM · Feature Engineering · Optuna · TreeSHAP

BIG DATA & DATA ENGINEERING

Apache Spark · Scala · PySpark · AWS EMR · Parquet · S3 Data Lakes · SQL (Window Func.) · DuckDB · Neo4j

CLOUD, MLOPS & SYSTEMS

AWS (EMR, S3, EC2) · Docker · Jenkins CI/CD · Bash / Linux · Git Workflows · REST APIs · SNORT IDS

FEATURED PROJECTS

Airline Demand Forecasting — Amadeus Production

Re-engineered distributed ETL & demand forecasting pipeline on **AWS EMR (Spark/Scala)**. Replaced PySpark bottlenecks with native Scala Catalyst passes, delivering verified **35% latency reduction** over multi-GB Parquet datasets.

Financial Time-Series Engine — Open Source / Quant

Automated multi-asset pipeline (CCXT, yfinance, MT5) with **73+ engineered alpha features**. Validated gradient boosting (XGBoost) and LSTMs with **strict purged walk-forward cross-validation (0 lookahead leakage)**.

RecSys & Ranking Engine — Applied ML

High-dimensional collaborative & content-based ranking engine. Optimized via **Bayesian search (Optuna)** with **TreeSHAP feature explainability** to mitigate cold-start drop-off.

SNORT Network IDS Optimization — Cybersecurity

Real-time traffic telemetry analysis across TCP/IP, UDP, ICMP protocols. Optimized heuristic inspection rules, reducing **false positive alarms by ~20%**.

CERTIFICATIONS

- **AWS Solutions Architect – Associate:** In Progress (Target Q3 2026)
- **Neo4j Certified Professional:** Neo4j Graph Academy (2024)
- **Certified Ethical Hacker (CEH):** EC-Council (2019)
- **Fortinet NSE 1, 2, 3:** Certified (Valid through June 2026)

LANGUAGES

- **English:** Native / Bilingual (C2)
- **French:** Professional Working Proficiency (B1/B2)
- **Hindi & Bengali:** Native / Bilingual